CS-471: Machine Learning

Week 4

Instructor: Dr. Hashir Kiani

Classroom Etiquette and Participation Guidelines

- Treat everyone with respect and courtesy at all times
- Speak only when it is your turn and raise your hand before contributing to the discussion
- Do not interrupt others while they are speaking
- Join the class on time
- Keep mobile phones on silent during the lecture

Lecture Outline

- Recap of Previous Lecture
- Logistic Regression
- Logistic Regression Example
- Multiclass Classification

Recap of Previous Lecture

- What is the hypothesis class in logistic regression?
- What is sigmoid function?
- How do we interpret the output of logistic regression?
- What is the loss function in logistic regression?

Logistic Regression – Loss Function

- Logistic loss also known as Binary Cross Entropy Loss:

$$Loss\big(y, h(\boldsymbol{x})\big) = -ylog\big(h(\boldsymbol{x})\big) - (1-y)\log(1-h(\boldsymbol{x}))$$

- Training loss in this case:

$$TrainLoss(\boldsymbol{w}) = -\frac{1}{N}\sum_{i=1}^{N}[y_i log\big(h(\boldsymbol{x_i})\big) + (1-y_i)\log(1-h(\boldsymbol{x_i}))]$$

Logistic Regression – Optimization Algorithm

- Gradient Descent:

$$\boldsymbol{w}_{new} = \boldsymbol{w}_{old} - \alpha \nabla_{\boldsymbol{w}} TrainLoss(\boldsymbol{w})$$

$$\nabla_{\boldsymbol{w}} TrainLoss(\boldsymbol{w}) = \frac{1}{N} \sum_{i=1}^{N} [(h(\boldsymbol{x_i}) - y_i)\boldsymbol{x_i}]$$

$$\boldsymbol{w}_{new} = \boldsymbol{w}_{old} - \frac{\alpha}{N} \sum_{i=1}^{N} [(h(\boldsymbol{x_i}) - y_i)\boldsymbol{x_i}]$$

Linear Classification Example

- Training data:

x	y
$[0 \quad 2]$	1
$[-2 \quad 0]$	1
$[1 \quad -1]$	-1

Multi-class Classification – One vs Rest

- One-vs-Rest approach also known as One-vs-All
- Strategy for handling multiclass classification problems with binary classifiers.
- Original multiclass problem is decomposed into multiple binary classification problems.
- For a problem with N classes, N separate binary classifiers are trained.
- Each classifier is responsible for distinguishing between one of the classes and the rest of the classes combined.
- Each classifier is trained on the entire dataset.
- Prediction is the class for which the corresponding classifier has the highest confidence score or probability.

Multi-class Classification – One vs One

- Strategy for handling multiclass classification problems with binary classifiers.
- Original multiclass problem is decomposed into multiple binary classification problems.
- For a problem with N classes, N(N-1)/2 separate binary classifiers are trained.
- Each classifier is responsible for distinguishing between one class and a single other class.
- Each classifier is trained on a subset of entire dataset.
- Prediction is the class that has the maximum number of votes.

Questions?